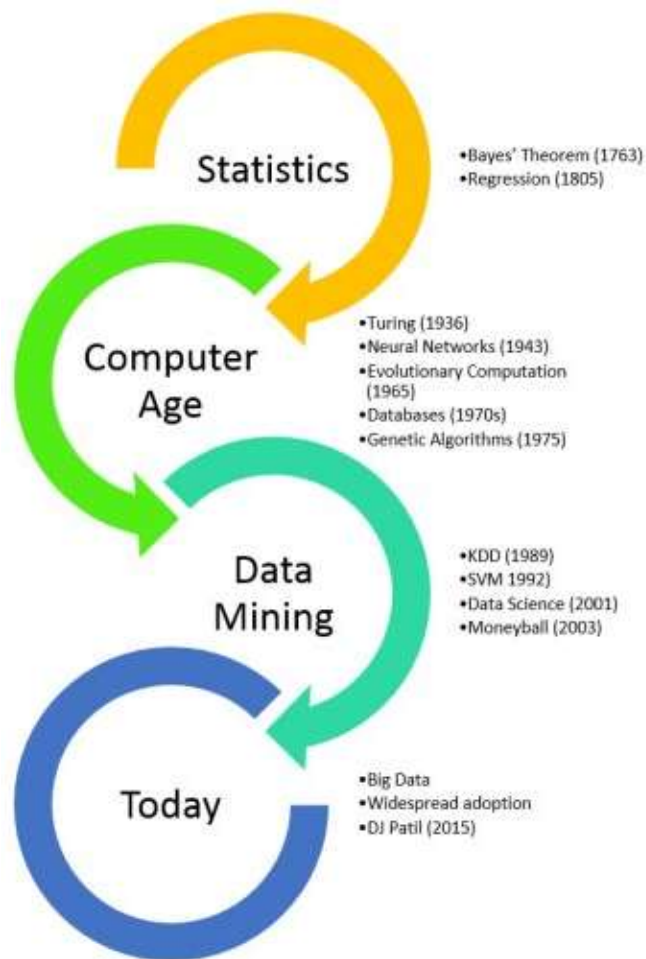


Unsupervised Clustering

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Evolutionary Step	Business Question	Enabling Technologies	Product Providers	Characteristics
Data Collection (1960s)	"What was my total revenue in the last five years?"	Computers, tapes, disks	IBM, CDC	Retrospective, static data delivery
Data Access (1980s)	"What were unit sales in New England last March?"	Relational databases (RDBMS), Structured Query Language (SQL), ODBC	Oracle, Sybase, Informix, IBM, Microsoft	Retrospective, dynamic data delivery at record level
Data Warehousing & Decision Support (1990s)	"What were unit sales in New England last March? Drill down to Boston."	On-line analytic processing (OLAP), multidimensional databases, data warehouses	Pilot, Comshare, Arbor, Cognos, Microstrategy	Retrospective, dynamic data delivery at multiple levels
Data Mining (Emerging Today)	"What's likely to happen to Boston unit sales next month? Why?"	Advanced algorithms, multiprocessor computers, massive databases	Pilot, Lockheed, IBM, SGI, numerous startups (nascent industry)	Prospective, proactive information delivery

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Syllabus

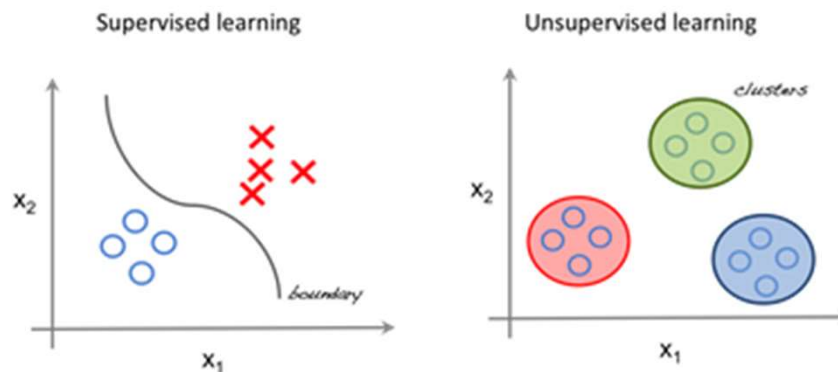
- Lecture 1 : Data Preprocessing
- Lecture 2 : Explanatory Data Analysis
- Lecture 3 : Feature Engineering (Feature Importance and Selection)
- Lecture 4 : Association Rule Learning
- **Lecture 5 : Unsupervised Clustering**
- **Lecture 6 : Unsupervised Clustering (cont.)**
- Lecture 7 : Anomaly and Outlier Detection
- Lecture 8 : Regression and Classification Learning
- Lecture 9 : Recommendation Learning
- Lecture 10 : Final Project Requirement

Outline

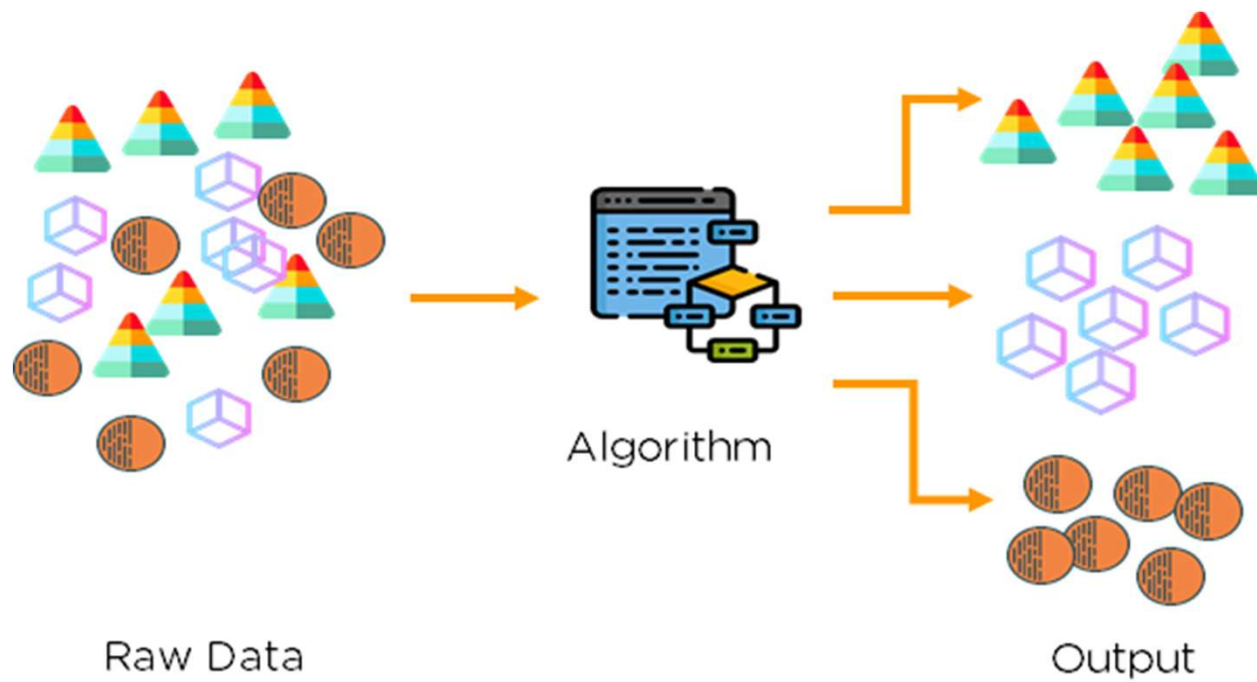
- Unsupervised learning:
 - K-means
 - Mean-shift
 - Gaussian Mixture Models
 - Hierarchy Clustering
 - Density-Based Spatial Clustering

Unsupervised learning

- **Supervised Learning** : our data had some target variables with specific values that we used to train our models
- **Unsupervised Learning** : However, when dealing with real-world problems, most of the time, data will not come with predefined labels, so we will want to develop machine learning models that can classify correctly this data, by finding by themselves some commonality in the features, that will be used to predict the classes on new data.

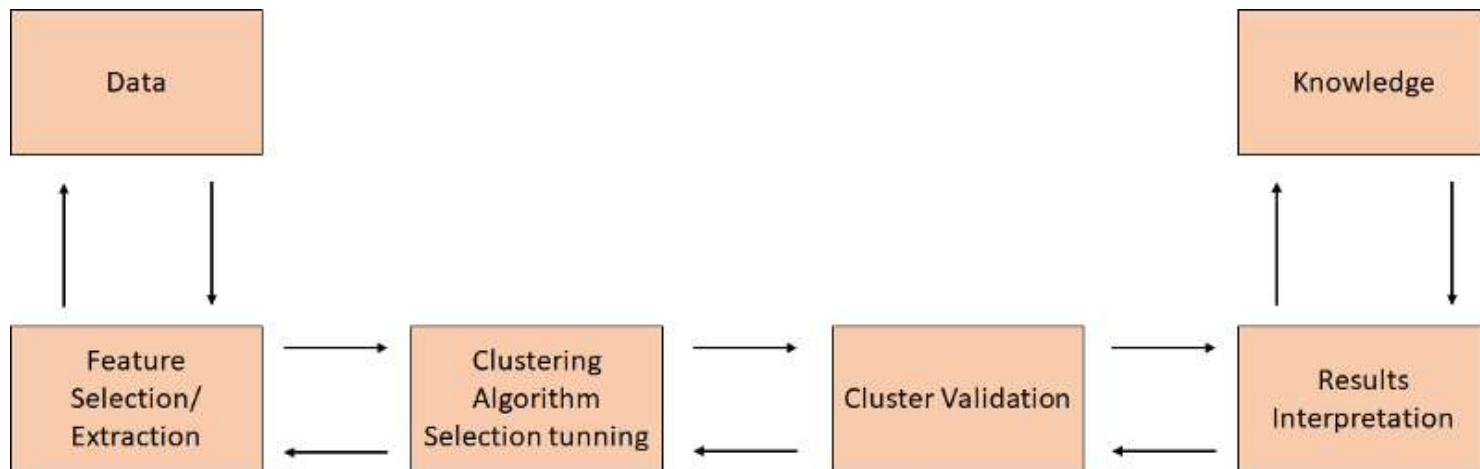


Unsupervised learning



Unsupervised learning

- **Unsupervised Learning Analysis Process**
- The overall process that we will follow when developing an unsupervised learning model can be summarized in the following chart:



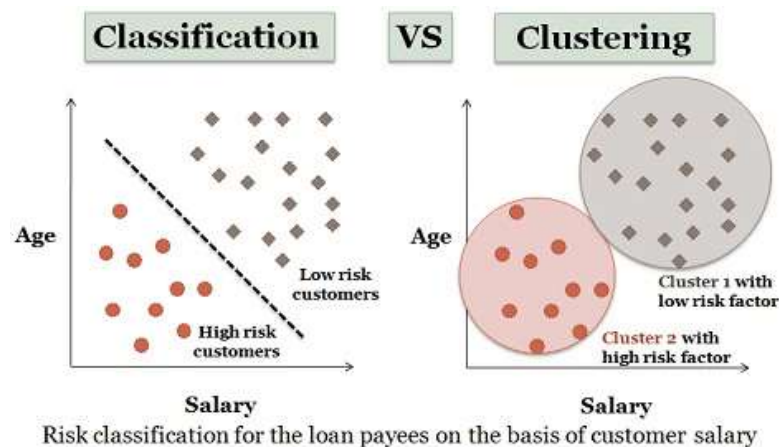
Unsupervised learning

- **Unsupervised learning main applications are:**
 - Segmenting datasets by some shared attributes.
 - Detecting anomalies that do not fit to any group.
 - Simplify datasets by aggregating variables with similar attributes.
- In summary, the main goal is to study the intrinsic (and commonly hidden) structure of the data.
- This techniques can be condensed in two main types of problems that unsupervised learning tries to solve. This problems are:
 - Clustering
 - Dimensionality Reduction

Unsupervised learning

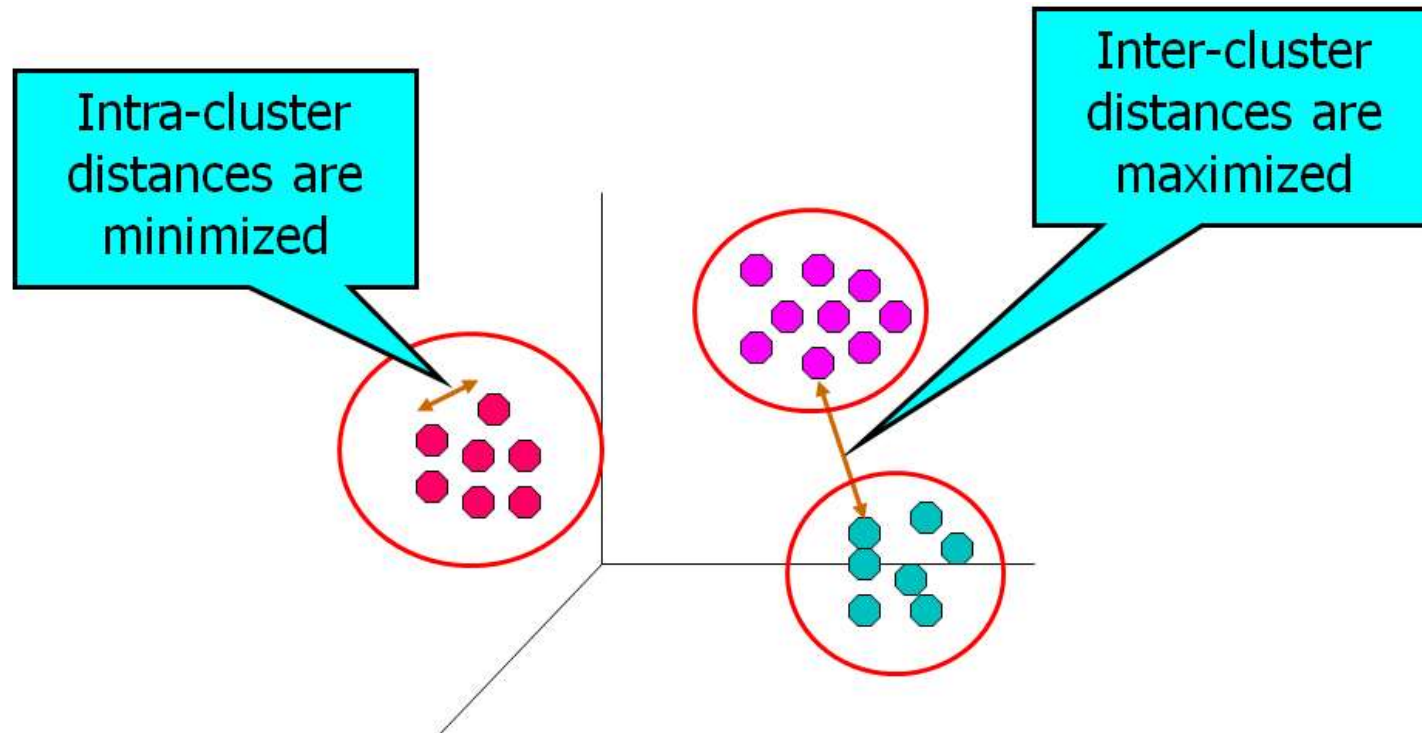
- **Clustering Analysis**

- In basic terms, the objective of clustering is to find different groups within the elements in the data. To do so, clustering algorithms find the structure in the data so that elements of the same cluster (or group) are more similar to each other than to those from different clusters.



Unsupervised learning

- **Clustering Analysis**



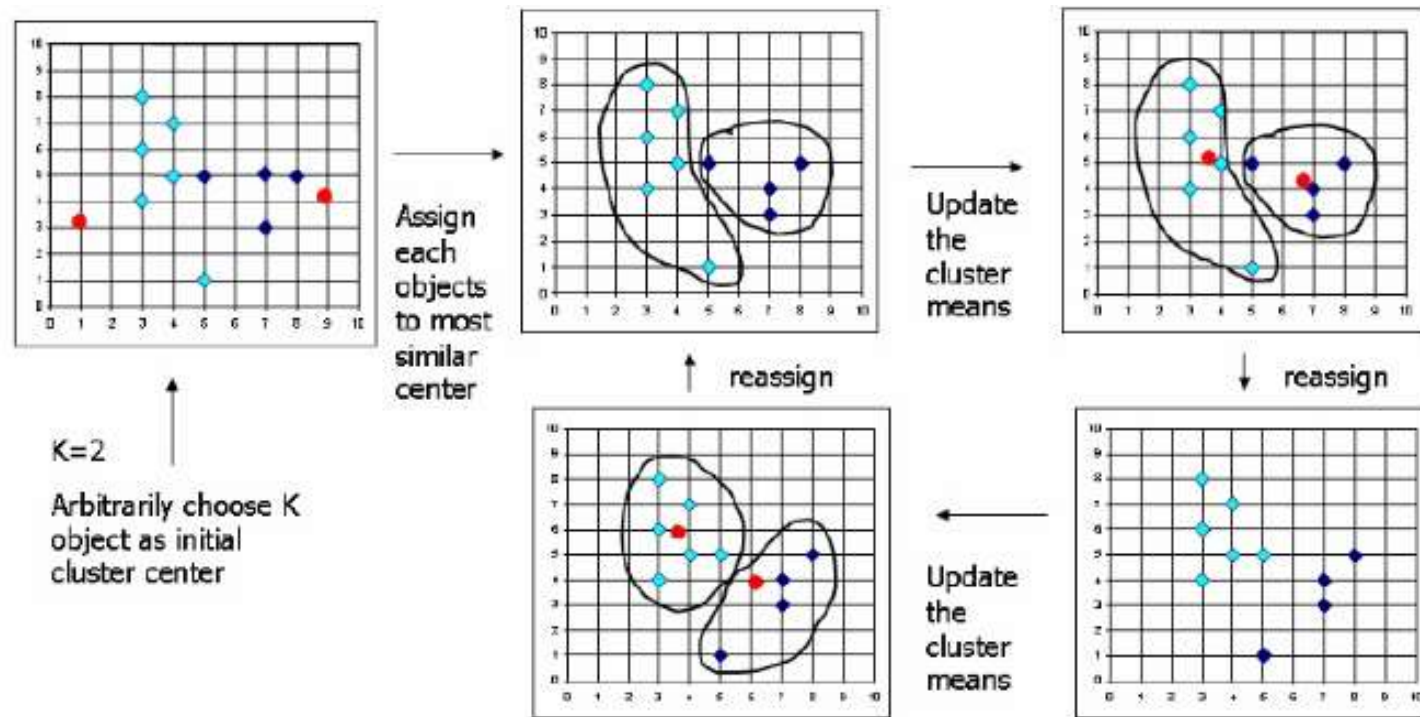
Unsupervised learning

- **Clustering Analysis**
- Some of the most common clustering algorithms, and the ones that will be explored throughout the article, are:
 - K-Means
 - Hierarchical Clustering
 - Density-Based Spatial Clustering of Applications with Noise (DBSCAN)
 - Gaussian Mixture Models (GMM)

K-Means Clustering

- The K-Means algorithm aims to find and group into classes the data points that have high similarity between them. In the terms of the algorithm, this similarity is understood as the opposite of the distance between datapoints.
- The closer the data points are, the more similar and more likely to belong to the same cluster they will be.

K-Means Clustering



K-Means Clustering

- **Key Concepts**
- Squared Euclidean Distance
- The most commonly used distance in K-Means is the squared Euclidean distance. An example of this distance between two points x and y in m -dimensional space is:
- Here, j is the j th dimension (or feature column) of the sample points x and y .

$$d(\mathbf{x}, \mathbf{y})^2 = \sum_{j=1}^m (x_j - y_j)^2 = \|\mathbf{x} - \mathbf{y}\|_2^2$$

K-Means Clustering

- **Key Concepts**
- Cluster Inertia
- Cluster inertia is the name given to the Sum of Squared Errors within the clustering context, and is represented as follows:
- Where $\mu(j)$ is the centroid for cluster j , and $w(i,j)$ is 1 if the sample $x(i)$ is in cluster j and 0 otherwise.

$$SSE = \sum_{i=1}^n \sum_{j=1}^k w^{(i,j)} \left\| x^{(i)} - \mu^{(j)} \right\|_2^2$$

Algorithm Steps

- First, we need to choose k , the number of clusters that we want to be found.
- Then, the algorithm will select randomly the centroids of each cluster.
- It will be assigned each datapoint to the closest centroid (using euclidean distance).
- It will be computed the cluster inertia.
- The new centroids will be calculated as the mean of the points that belong to the centroid of the previous step. In other words, by calculating the minimum quadratic error of the datapoints to the center of each cluster, moving the center towards that point
- Back to step 3.

K-Means Clustering

- **K-Means Hyperparameters**

- Number of clusters: The number of clusters and centroids to generate.
- Maximum iterations: Of the algorithm for a single run.
- Number initial: The number of times the algorithm will be run with different centroid seeds. The final result will be the best output of the number defined of consecutive runs, in terms of inertia.

- **Challenges of K-Means**

- The output for any fixed training set won't be always the same, because the initial centroids are set randomly and that will influence the whole algorithm process.
- As stated before, due to the nature of Euclidean distance, it is not a suitable algorithm when dealing with clusters that adopt non-spherical shapes.

K-Means Clustering

- **Points to be Considered When Applying K-Means**
- Features must be measured on the same scale, so it may be necessary to perform z-score standardization or max-min scaling.
- When dealing with categorical data, we will use the get dummies function.
- Exploratory Data Analysis (EDA) is very helpful to have an overview of the data and determine if K-Means is the most appropriate algorithm.
- The minibatch method is very useful when there is a large number of columns, however, it is less accurate.

K-Means Clustering

- **How to Choose the Right K Number**
- Choosing the right number of clusters is one of the key points of the K-Means algorithm. To find this number there are some methods:
 - Field knowledge
 - Bussiness decision
 - Elbow Method
- As being aligned with the motivation and nature of Data Science, the elbow mehtod is the prefered option as it relies on an analytical method backed with data, to make a decision.

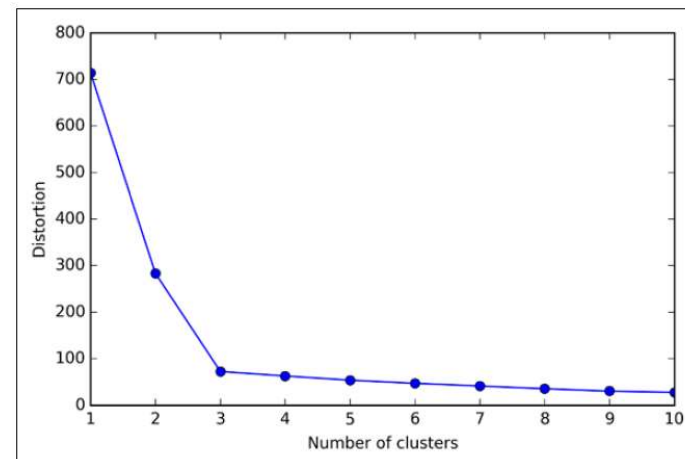
K-Means Clustering

- **Elbow Method**

- The elbow method is used for determining the correct number of clusters in a dataset. It works by plotting the ascending values of K versus the total error obtained when using that K.

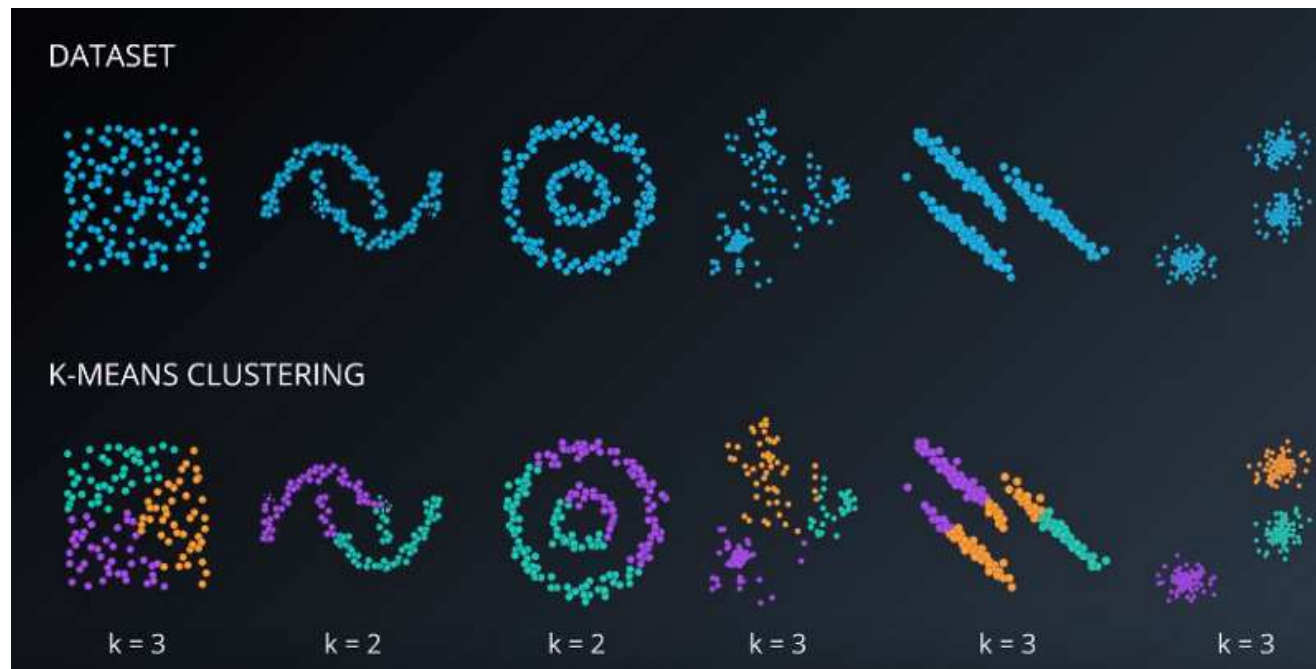
- The goal is to find the k that for each cluster will not rise significantly the variance
In this case, we will choose the k=3, where the elbow is located.

$$\% \text{ Variance} = \frac{\text{Variance between groups}}{\text{Total variance}}$$



K-Means Clustering

K-Means Limitations

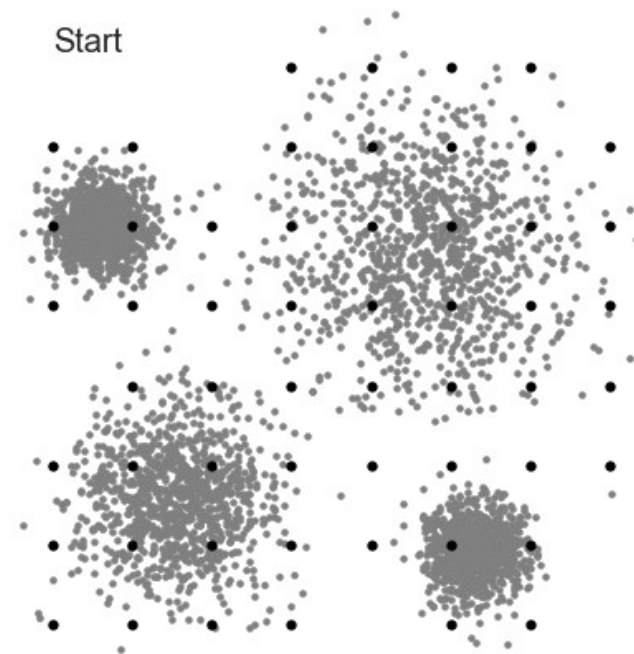
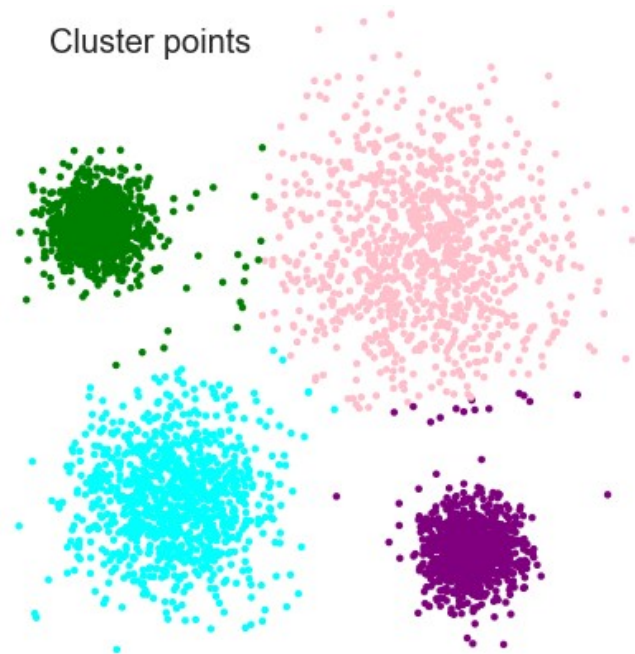


It is quite common to take the K-Means algorithm as a benchmark to evaluate the performance of other clustering methods.

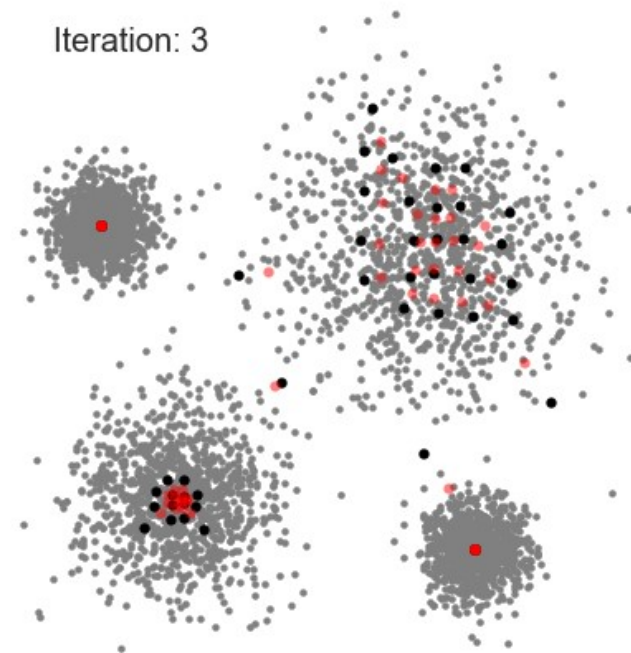
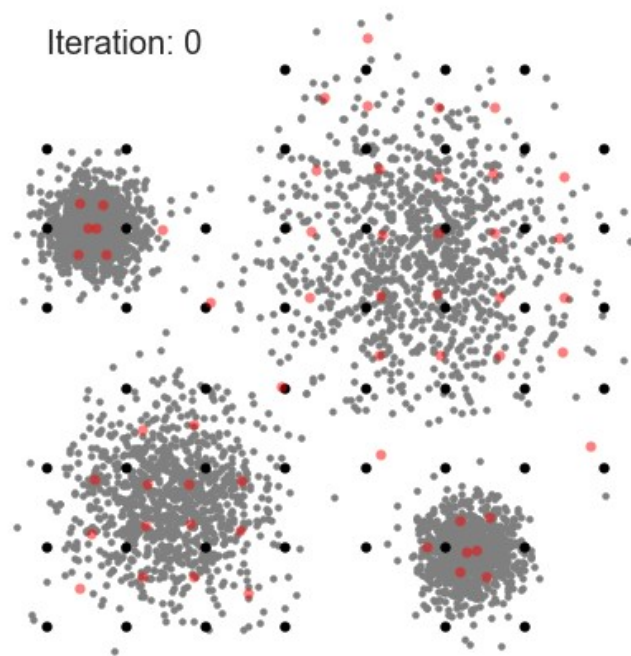
Mean-Shift Clustering

- Mean shift clustering is a sliding-window-based algorithm that attempts to find dense areas of data points. It is a centroid-based algorithm meaning that the goal is to locate the center points of each group/class, which works by updating candidates for center points to be the mean of the points within the sliding-window.
- These candidate windows are then filtered in a post-processing stage to eliminate near-duplicates, forming the final set of center points and their corresponding groups. Check out the graphic below for an illustration

Mean-Shift Clustering

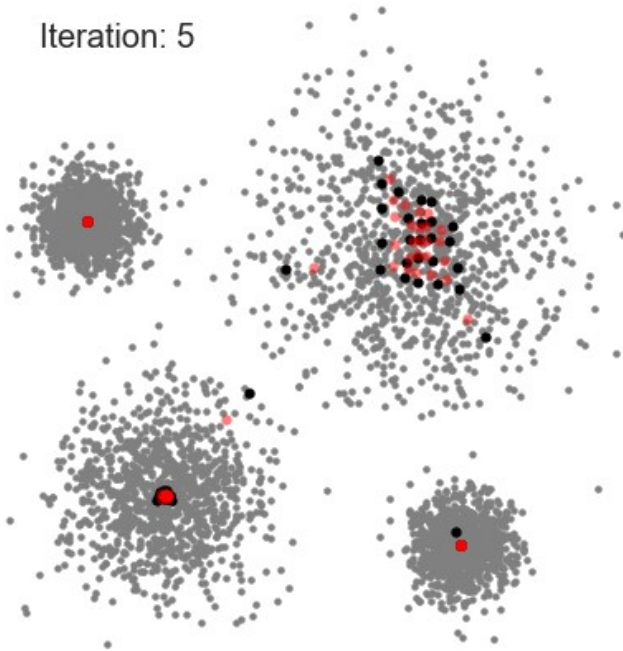


Mean-Shift Clustering

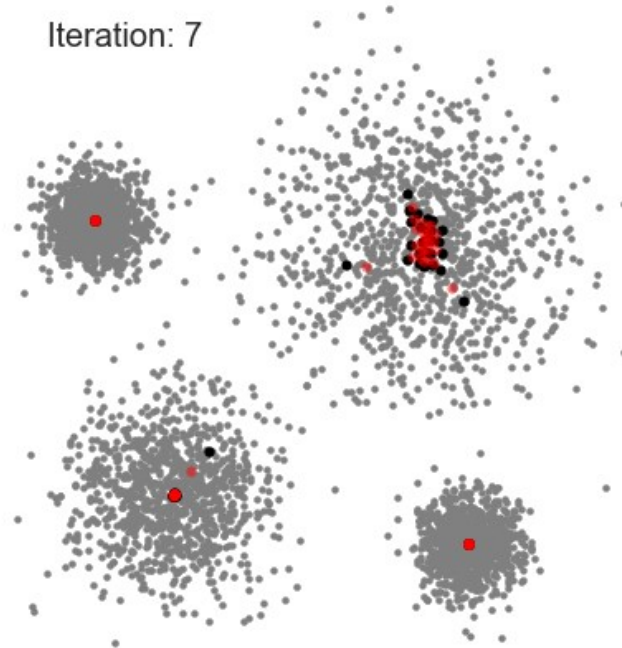


Mean-Shift Clustering

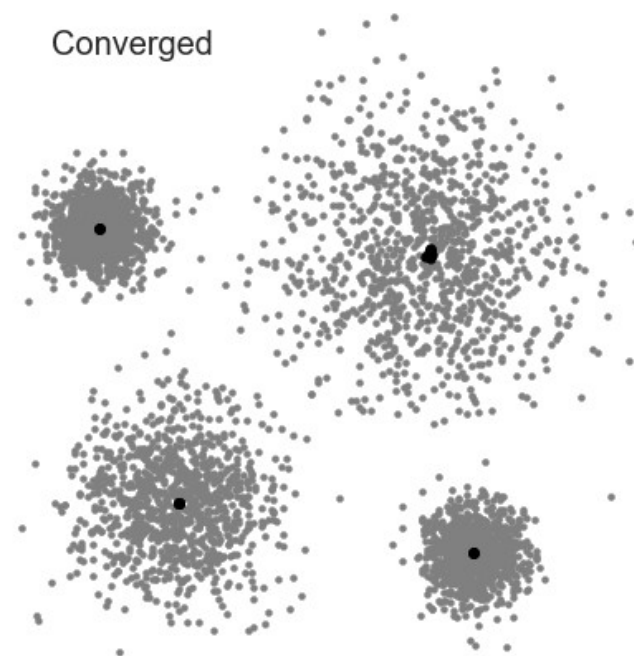
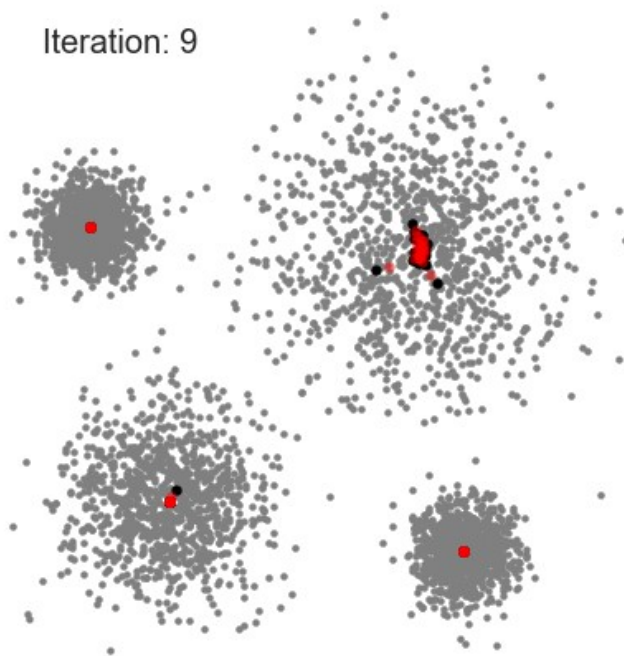
Iteration: 5



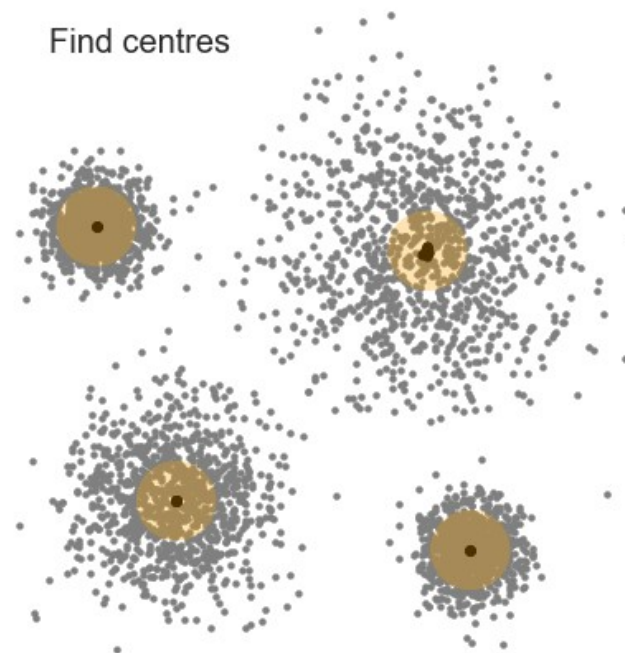
Iteration: 7



Mean-Shift Clustering



Mean-Shift Clustering



Mean-Shift Clustering

- 1. To explain mean-shift we will consider a set of points in two-dimensional space like the above illustration. We begin with a circular sliding window centered at a point C (randomly selected) and having radius r as the kernel. Mean shift is a hill climbing algorithm which involves shifting this kernel iteratively to a higher density region on each step until convergence.
- 2. At every iteration the sliding window is shifted towards regions of higher density by shifting the center point to the mean of the points within the window (hence the name). The density within the sliding window is proportional to the number of points inside it. Naturally, by shifting to the mean of the points in the window it will gradually move towards areas of higher point density.

Mean-Shift Clustering

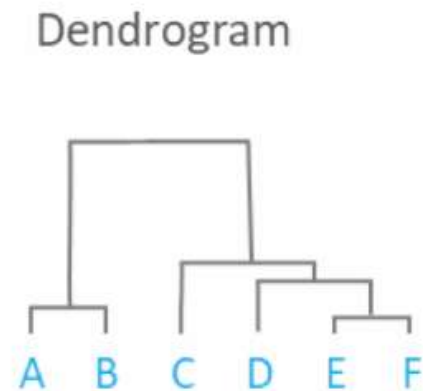
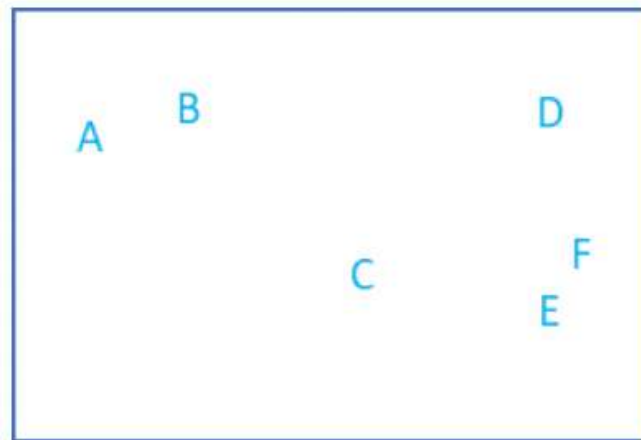
- 3. We continue shifting the sliding window according to the mean until there is no direction at which a shift can accommodate more points inside the kernel. Check out the graphic above; we keep moving the circle until we no longer are increasing the density (i.e number of points in the window).
- 4. This process of steps 1 to 3 is done with many sliding windows until all points lie within a window. When multiple sliding windows overlap the window containing the most points is preserved. The data points are then clustered according to the sliding window in which they reside

Mean-Shift Clustering

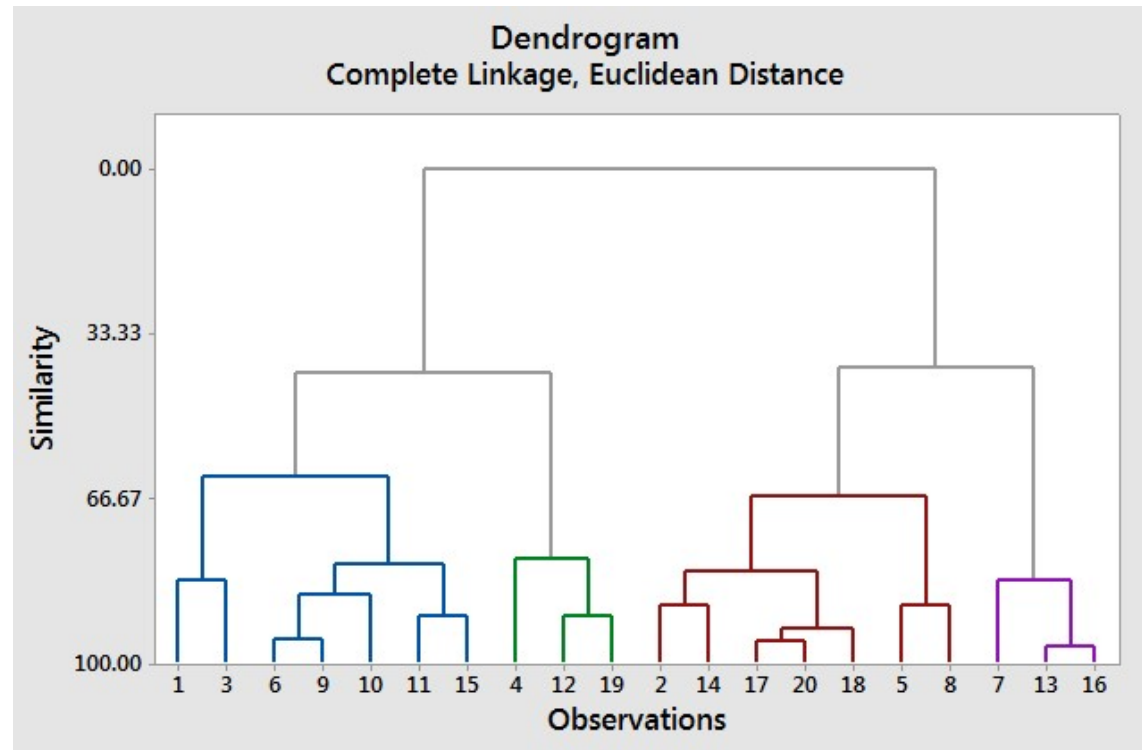
- In contrast to K-means clustering there is no need to select the number of clusters as mean-shift automatically discovers this. That's a massive advantage. The fact that the cluster centers converge towards the points of maximum density is also quite desirable as it is quite intuitive to understand and fits well in a naturally data-driven sense. The drawback is that the selection of the window size/radius " r " can be non-trivial.

Hierarchical Clustering

- Hierarchical clustering is an alternative to prototype-based clustering algorithms. The main advantage of Hierarchical clustering is that we do not need to specify the number of clusters, it will find it by itself.
- In addition, it enables the plotting of dendrograms. Dendrograms are visualizations of a binary hierarchical clustering.



Hierarchichal Clustering



Observations that fuse at the bottom are similar while those that are at the top are quite different. With dendrograms, conclusions are made based on the location of the vertical axis rather than on the horizontal one.

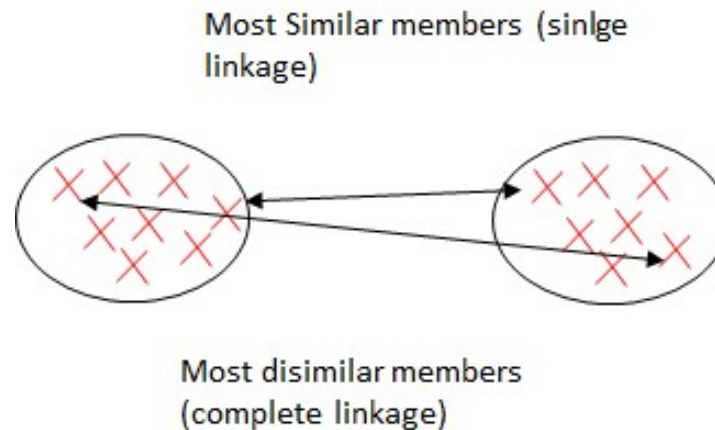
Hierarchichal Clustering

- **Kinds of Hierarchichal Clustering**

- There are two approaches to this type of clustering: Agglomerative and divisive.
- Divisive: this method starts by englobing all datapoints in one single cluster. Then, it will split the cluster iteratively into smaller ones until each one of them contains only one sample.
- Agglomerative: this method starts with each sample being a different cluster and then merging them by the ones that are closer from each other until there is only one cluster.

Hierarchichal Clustering

- **Single Linkage & Complete Linkage**
- These are the most common algorithms used for agglomerative hierarchichal clustering.



Hierarchichal Clustering

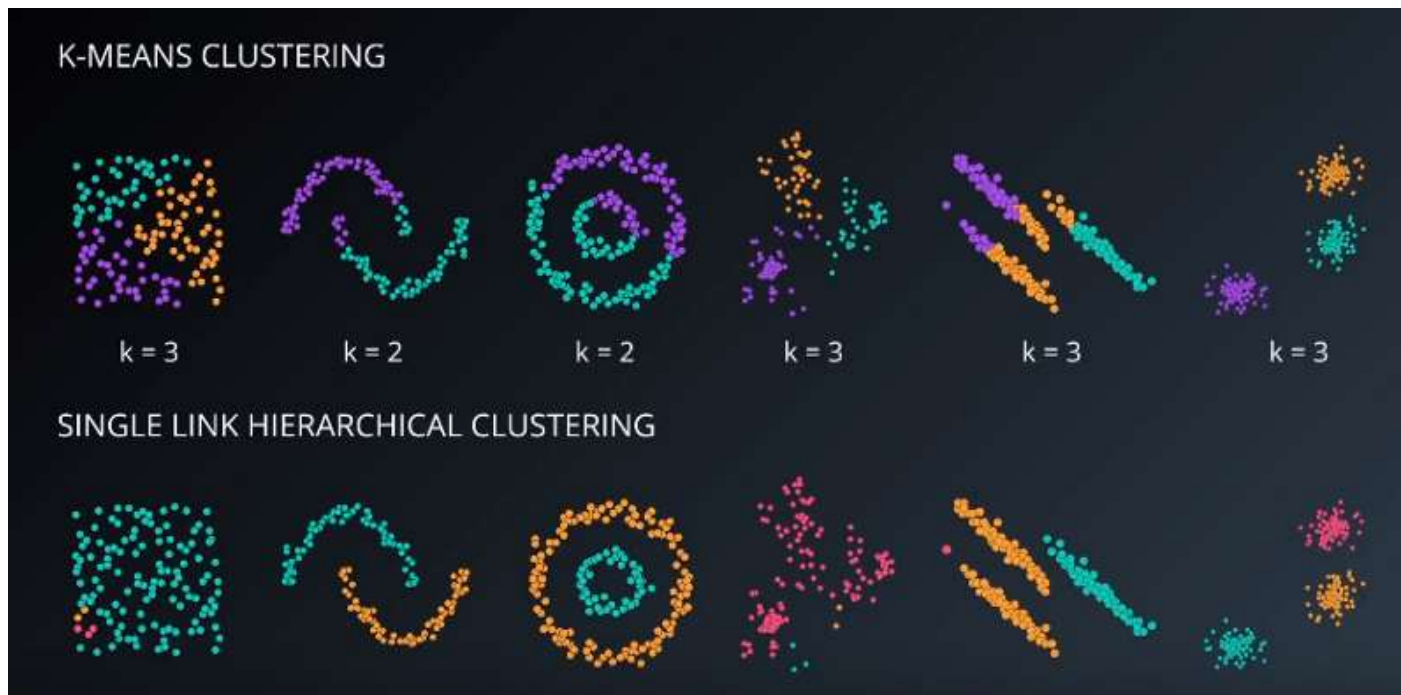
- **Single Linkage**

- As being an agglomerative algorithm, single linkage starts by assuming that each sample point is a cluster. Then, it computes the distances between the most similar members for each pair of clusters and merge the two clusters for which the distance between the most similar members is the smallest.

- **Complete Linkage**

- Although being similar to its brother (single linkage) its philosophy is esactly the opposite, it compares the most dissimilar datapoints of a pair of clusters to perform the merge.

Hierarchical Clustering



Hierarchichal Clustering

- **Advantages of Hierarchichal Clustering**

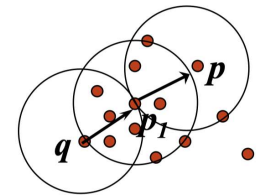
- The resulting hierarchichal representations can be very informative.
- Dendograms provide an interesting and informative way of visualization.
- They are specially powerful when the dataset comtains real hierarchichal relationships.

- **Disadvantages of Hierarchichal Clustering**

- They are very sensitive to outliers and, in their presence, the model performance decreases significantly.
- They are very expensive, computationally speaking.

Density-Based Spatial Clustering of Applications with Noise (DBSCAN)

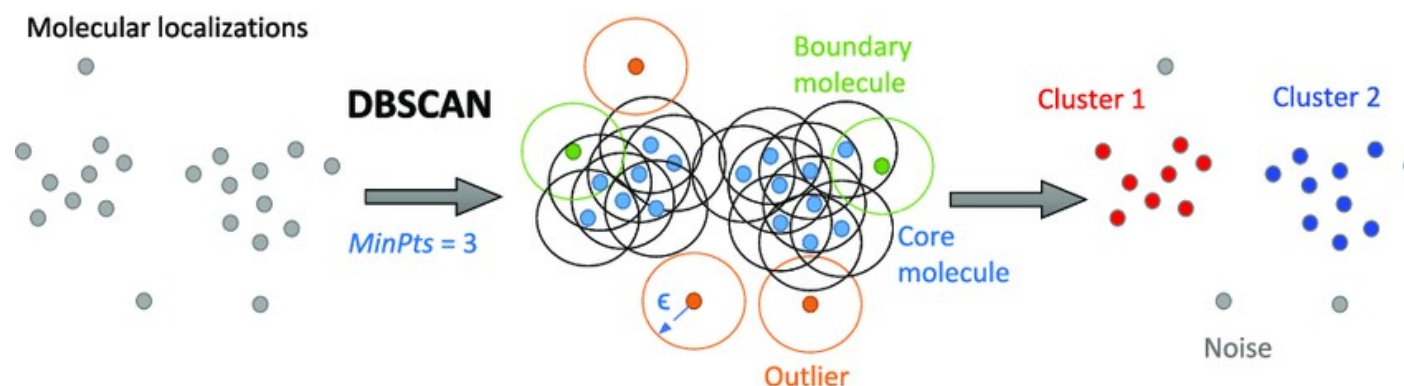
- Density-Based Spatial Clustering of Applications with Noise, or DBSCAN, is another clustering algorithm specially useful to correctly identify noise in data.
- **DBSCAN Assigning Criteria**
 - It is based on a number of points with a specified radius ϵ and there is a special label assigned to each datapoint. The process of assigning this label is the following:
 - It is a specified number (MinPts) of neighbour points. A core point will be assigned if there is this MinPts number of points that fall in the ϵ radius.
 - A border point will fall in the ϵ radius of a core point, but will have less neighbors than the MinPts number.
 - Every other point will be noise points.
- Density-reachable:
 - A point p is **density-reachable** from a point q w.r.t. Eps , $MinPts$ if there is a chain of points $p_1, \dots, p_n, p_1 = q, p_n = p$ such that p_{i+1} is directly density-reachable from p_i



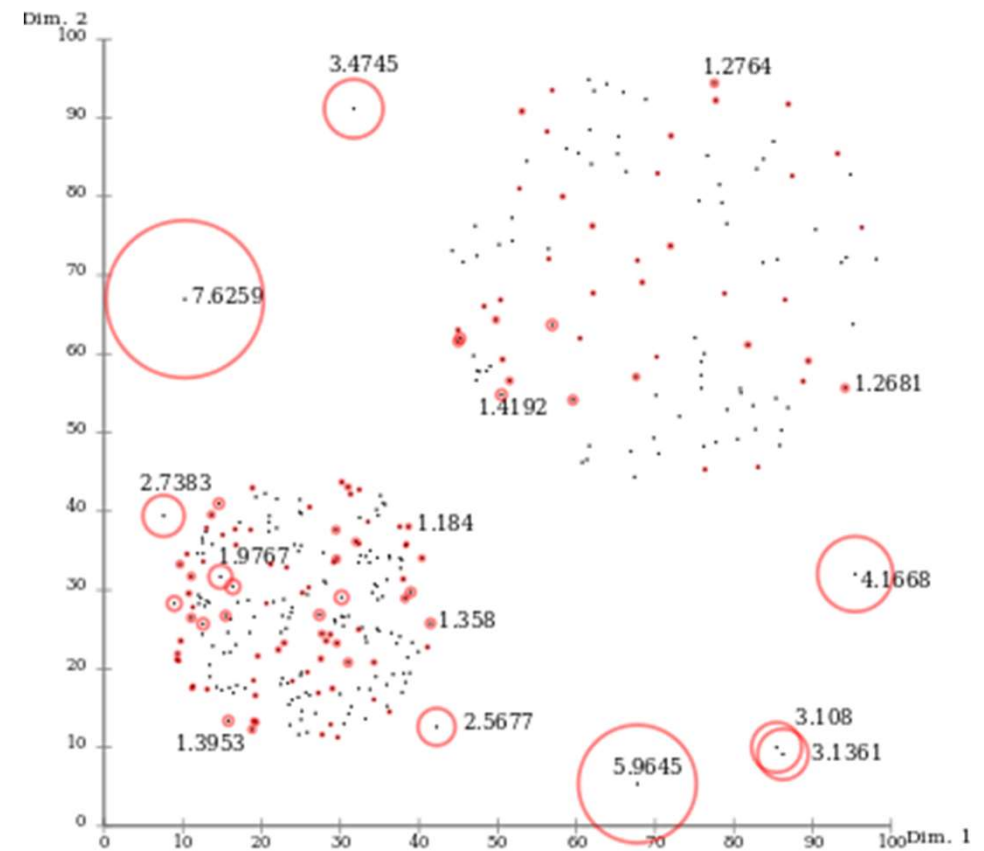
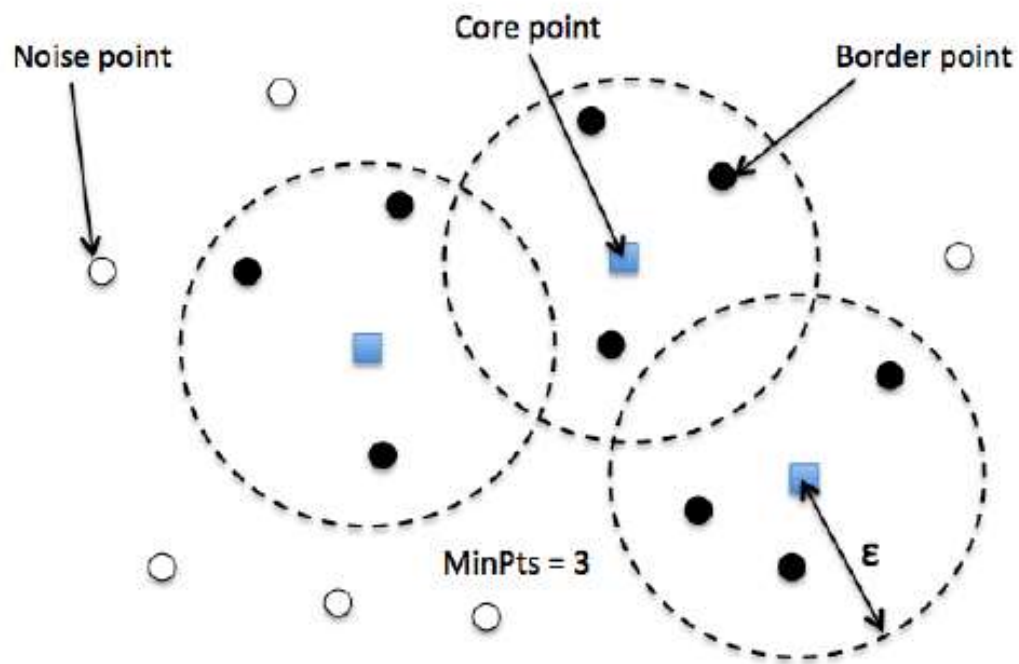
Density-Based Spatial Clustering of Applications with Noise (DBSCAN)

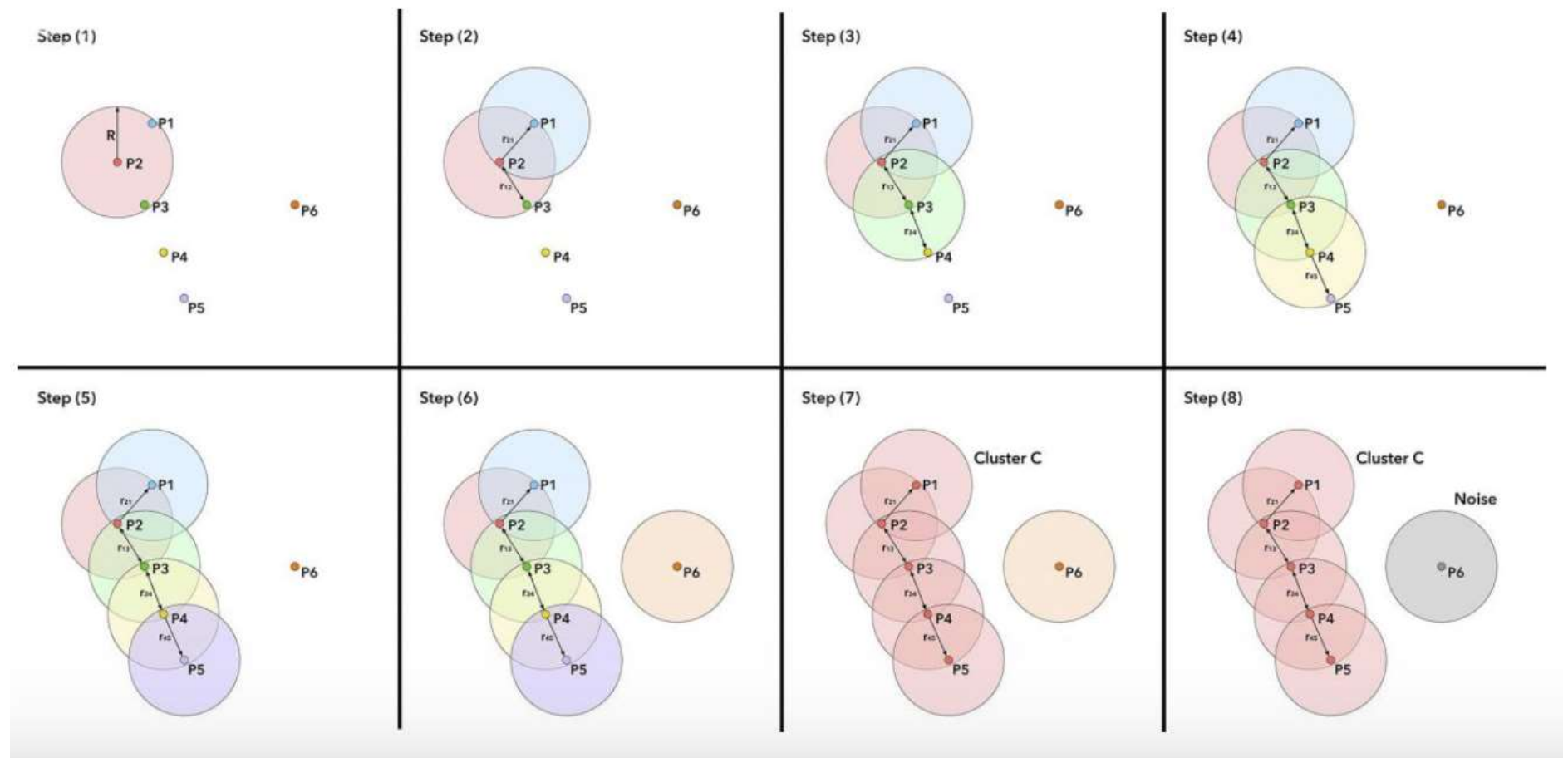
- **DBSCAN Algorithm**

- The algorithm follows the logic:
- Identify a core point and make a group for each one, or for each connected group of core points (if they satisfy the criteria to be core point).
- Identify and assign border points to their respective core points.
- The following figure summarize very well this process and the commented notation.



Density-Based Spatial Clustering of Applications with Noise (DBSCAN)





Density-Based Spatial Clustering of Applications with Noise (DBSCAN)

- **DBSCAN Advantages**

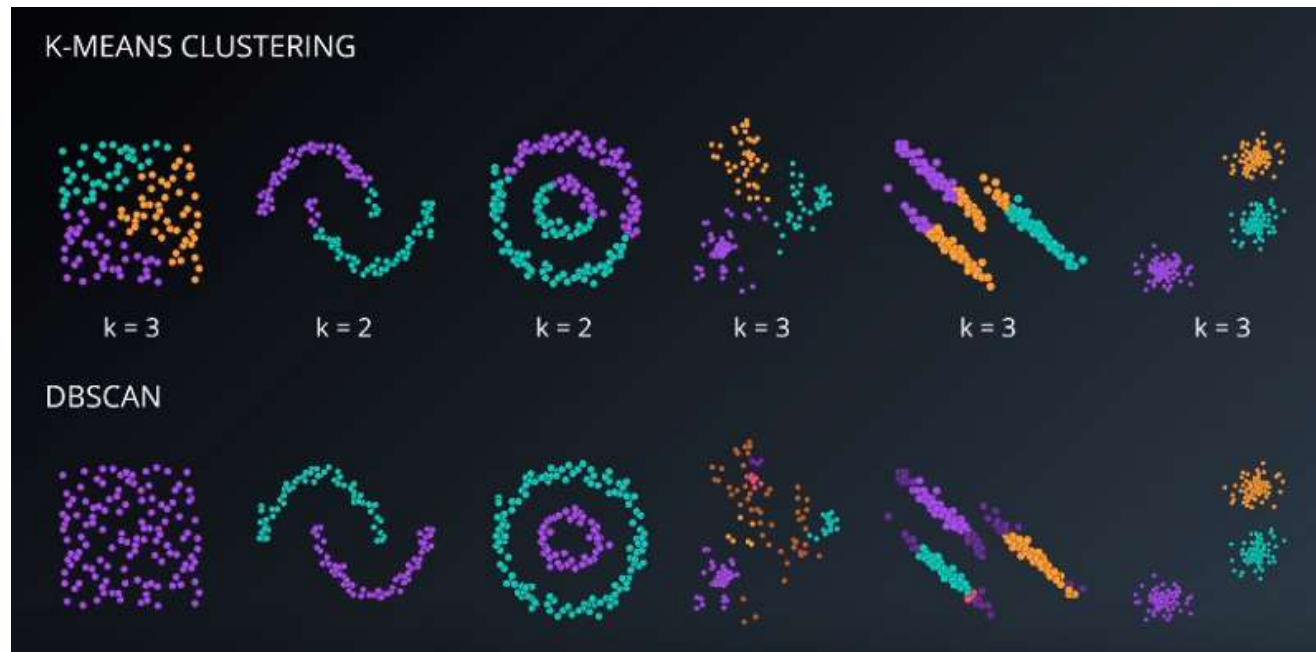
- We do not need to specify the number of clusters.
- There is high flexibility in the shapes and sizes that the clusters may adopt.
- It is very useful to identify and deal with noise data and outliers.

- **DBSCAN Disadvantages**

- It faces difficulties when dealing with boirder points that are reachable by two clusters.
- It doesn't find well clusters of varying densities.

Density-Based Spatial Clustering of Applications with Noise (DBSCAN)

- **DBSCAN vs K-Means Clustering**



Gaussian Mixture Models (GMM)

- Gaussian Mixture Models are probabilistic models that assume that all samples are generated from a mix of a finite number of Gaussian distribution with unknown parameters.
- It belongs to the group of soft clustering algorithms in which every data point will belong to every cluster existing in the dataset, but with different levels of membership to each cluster. This membership is assigned as the probability of belonging to a certain cluster, ranging from 0 to 1.

Gaussian Mixture Models (GMM)

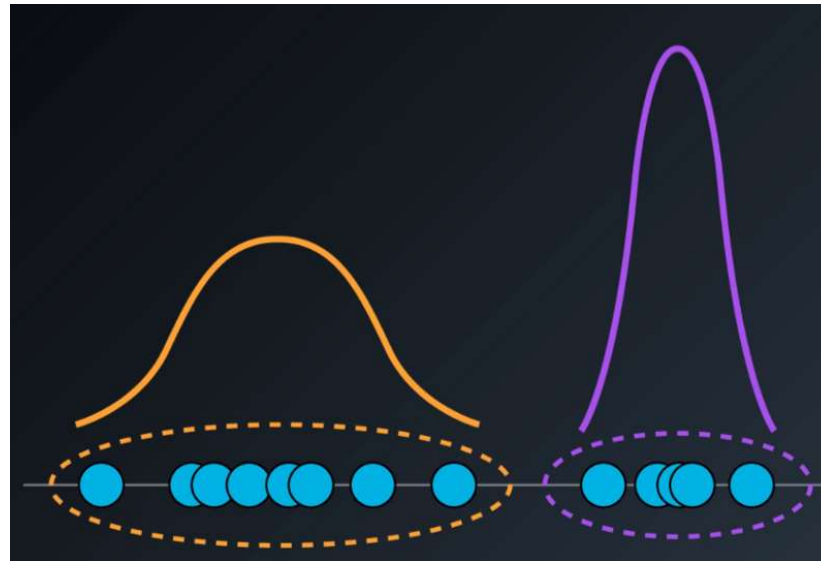
- For example, the highlighted point will belong to clusters A and B simultaneously, but with higher membership to the group A, due to its closeness to it.

-



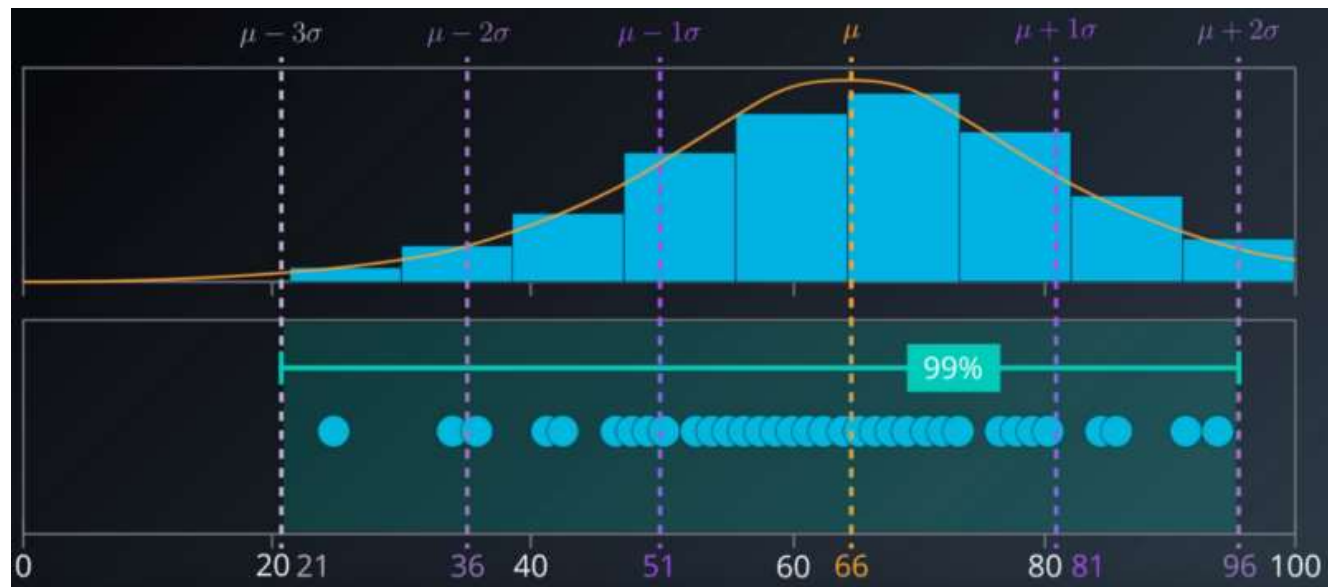
Gaussian Mixture Models (GMM)

- GMM is one of the most advanced clustering methods that we will study in this series, it assumes that each cluster follows a probabilistic distribution that can be Gaussian or Normal. It is a generalization of K-Means clustering that includes information about the covariance structure of the data as well as the centers of the latent Gaussians.



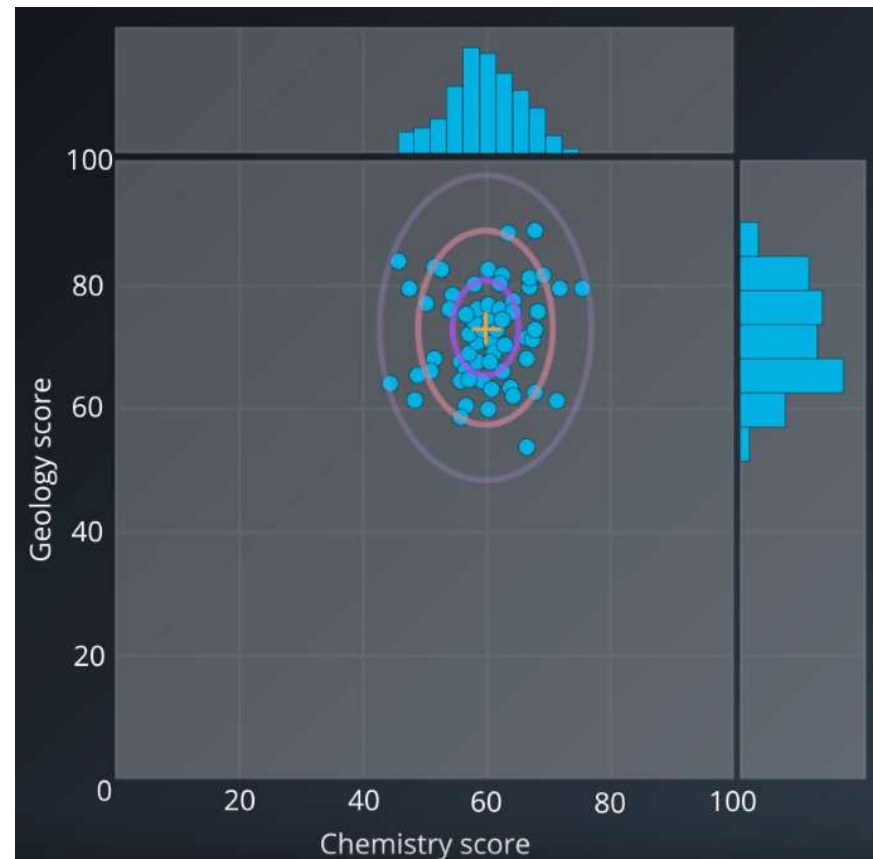
Gaussian Mixture Models (GMM)

- **GMM Distribution in One Dimension**
- The GMM will search for gaussian distributions in the dataset and mixture them.



Gaussian Mixture Models (GMM)

- **GMM in Two Dimensions**
- When having multivariate distributions as the following one, the mean centre would be $\mu + \sigma$, for each axis of the de dataset distribution.



Gaussian Mixture Models (GMM)

- **GMM Algorithm**
- It is an expectation-maximization algorithm which process could be summarize as follows:

1. Initialize K Gaussian distributions. It does this with the μ (mean) and σ (standard deviation) values. They can be taken from the dataset (naive method) or by applying K-Means.
2. Soft cluster the data: this is the 'Expectation' phase in which all datapoints will be assigned to every cluster with their respective level of membership.
3. Re-estimate the gaussians: this is the 'Maximization' phase in which the expectations are checked and they are used to calculate new parameters for the gaussians: new μ and σ .
4. Evaluate the log-likelihood of the data to check for convergence. The higher the log-likelihood is, the more probable is that the mixture of the model we created is likely to fit our dataset. So, this is the function to maximize.
5. Repeat from step 2 until convergence.

Gaussian Mixture Models (GMM)

- **GMM Advantages**

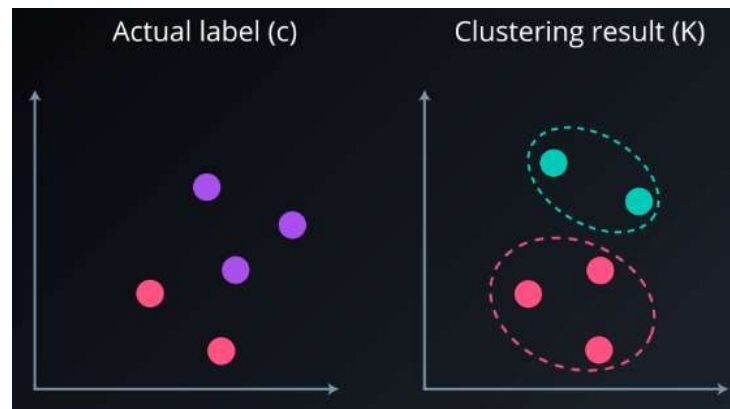
- It is a soft-clustering method, which assign sample memberships to multiple clusters. This characteristic makes it the fastest algorithm to learn mixture models
- There is high flexibility in the number and shape of the clusters.

- **GMM Disadvantages**

- It is very sensitive to the initial values which will condition greatly its performance.
- GMM may converge to a local minimum, which would be a sub-optimal solution.
- When having insufficient points per mixture, the algorithm diverges and finds solutions with infinite likelihood unless we regularize the covariances between the data points artificially.

Gaussian Mixture Models (GMM)

- **Clustering Validation**
- Clustering validation is the process of evaluating the result of a cluster objectively and quantitatively. We will do this validation by applying cluster validation indices. There are three main categories:
- **External Indices**
- These are scoring methods that we use if the original data was labelled, which is not the most frequent case in this kind of problems. We will match a clustering structure to information known beforehand.



Gaussian Mixture Models (GMM)

- The most used index is the Adjusted Rand index.
- Adjusted Rand Index (ARI) $\in [-1,1]$
- To understand it we should first define its components:

$$\text{Rand Index} = \frac{a + b}{\binom{n}{2}}$$

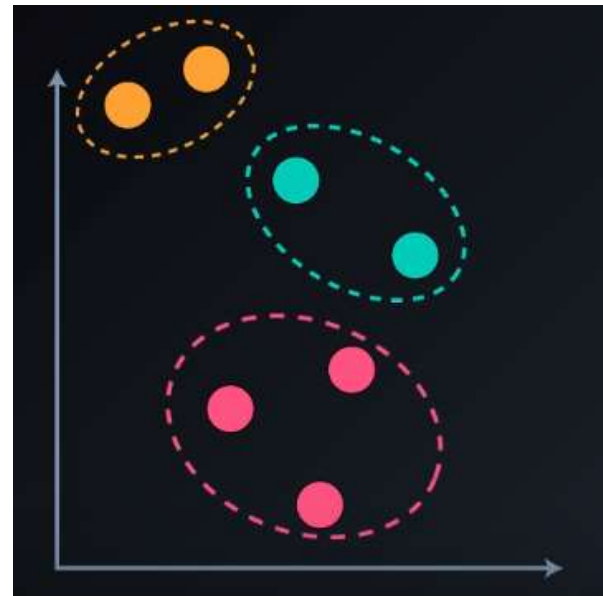
- a: is the number of points that are in the same cluster both in C and in K
- b: is the number of points that are in the different cluster both in C and in K.
- n = is the total number of samples

$$\text{ARI} = \frac{\text{RI} - \text{Expected Index}}{\text{Max(RI)} - \text{Expected Index}}$$

- The ARI can get values ranging from -1 to 1. The higher the value, the better it matches the original data.

Gaussian Mixture Models (GMM)

- **Internal Validation Indices**
- In unsupervised learning, we will work with unlabeled data and this is when internal indices are more useful.
- One of the most common indices is the Silhouette Coefficient.
- Silhouette Coefficient:
- There is a Silhouette Coefficient for each data point.



Gaussian Mixture Models (GMM)

- a = average distance to other sample i in the same cluster
- b = average distance to other sample i in closest neighbouring cluster

$$S_i = \frac{b_i - a_i}{\text{Max}(b_i - a_i)}$$

S_c = average of the sum of S_i for each point

- The Silhouette Coefficient (SC) can get values from -1 to 1. The higher the value, the better the K selected is. It penalized more if we surpass the ideal K than if we fall short.
- It is only suitable for certain algorithms such as K-Means and hierarchical clustering. It is not suitable to work with DBSCAN, we will use DBCV instead.



THANK YOU

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Science, University of Science, HCMC